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Handling machine control method, handling machine control system and control host

Abstract

A handling machine control method, a handling machine control system and a control host are provided. The method includes: detecting a handling instruction which instructs a handling of a first object; reading handling information of the first object according to the handling instruction; if the handling information of the first object is read, control, according to the handling information, a first handling machine to move to a first position and perform a first operation to lift up the first object at the first position; after the first object is lifted up, control the first handling machine to move to a second position and perform a second operation to put down the first object at the second position; and in response to that the first object is put down at the second position, updating the handling information of the first object.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 110143515, filed on Nov. 23, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

(2) This disclosure relates to a handling machine control method, a handling machine control system and a control host.

BACKGROUND

(3) As people's transaction patterns change and the demand for automated logistics management such as smart warehousing is growing, it is desirable for an increasing number of logistics companies to use automated handling devices to automatically complete the handling of goods and the loading and unloading of goods. However, the cargo to be transported and/or the pallets carrying the cargo are often arranged irregularly in the physical space. Therefore, it is not easy to complete the automated handling of goods through simple coordinate setting.

SUMMARY

(4) In view of this, the disclosure provides a handling machine control method, a handling machine control system and a control host, which can improve the efficiency of automated cargo handling.

(5) An embodiment of the disclosure provides a handling machine control method, which is adaptable for a handling machine control system. The handling machine control method includes: detecting a handling instruction which instructs a handling of a first object; reading handling information of the first object according to the handling instruction; if the handling information of the first object is read, controlling, according to the handling information of the first object, a first handling machine to move to a first position and perform a first operation to lift up the first object at the first position; after the first object is lifted up, controlling the first handling machine to move to a second position and perform a second operation to put down the first object at the second position; and in response to that the first object is put down at the second position, updating the handling information of the first object.

(6) An embodiment of the disclosure further provides a handling machine control system, which includes a control host and a first handling machine. The first handling machine is coupled to the control host. The control host is configured to: detect a handling instruction which instructs a handling of a first object; read handling information of the first object according to the handling instruction; if the handling information of the first object is read, control, according to the handling information of the first object, the first handling machine to move to a first position and perform a first operation to lift up the first object at the first position; after the first object is lifted up, control the first handling machine to move to a second position and perform a second operation to put

down the first object at the second position; and in response to that the first object is put down at the second position, update the handling information of the first object.

(7) An embodiment of the disclosure further provides a control host, which is adapted to control at least one handling machine. The control host includes a storage circuit and a processor. The storage circuit is configured for storing the handling information of the first object. The processor is coupled to the storage circuit. The processor is configured to: detect a handling instruction which instructs a handling of a first object; read handling information of the first object according to the handling instruction; if the handling information of the first object is read, control, according to the handling information of the first object, a first handling machine among the at least one handling machine to move to a first position and perform a first operation to lift up the first object at the first position; after the first object is lifted up, control the first handling machine to move to a second position and perform a second operation to put down the first object at the second position; and in response to that the first object is put down at the second position, update the handling information of the first object.

(8) Based on the above, after detecting the handling instruction, the system can try to read the handling information of the first object. According to the handling information of the first object, the system can control the first handling machine to move to the first position and perform the first operation to lift up the first object at the first position. After the first object is lifted up, the system can control the first handling machine to move to the second position and perform the second operation to put down the first object at the second position. In addition, in response to the first object being put down at the second position, the handling information of the first object can be updated. In this way, the efficiency of automated cargo handling can be effectively improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a handling machine control system according to an embodiment of the disclosure.

(2) FIG. 2 is a functional block diagram of a control host according to an embodiment of the disclosure.

(3) FIG. 3 is a schematic diagram of adjusting the posture of the first handling machine according to an embodiment of the disclosure.

(4) FIG. 4 is a schematic diagram of image recognition according to an embodiment of the disclosure.

(5) FIG. 5 is a flowchart of a handling machine control method according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

(6) FIG. 1 is a schematic diagram of a handling machine control system according to an embodiment of the disclosure. Please refer to FIG. 1, the system (also referred to as the handling machine control system) **10** includes a control host **11** and handling machines **12(1)** to **12(n)**. The control host **11** can be configured to manage and control the handling machines **12(1)** to **12(n)**. For example, the control host **11** can monitor the position, movement trajectory and posture of the handling machines **12(1)** to **12(n)** and control the handling machines **12(1)** to **12(n)** to move and/or handle specific objects. The total number of control host **11** and the total number of handling machines **12(1)** to **12(n)** can be adjusted according to practical requirements, and this disclosure is not limited thereto.

(7) In an embodiment, the control host **11** is independently arranged outside the handling machines **12(1)** to **12(n)** and can communicate with the handling machines **12(1)** to **12(n)** wirelessly. For example, the control host **11** may include a desktop computer, a notebook computer, a smart phone,

a tablet computer, an industrial computer, a server, or other types of computer devices with data transmission and processing functions. Alternatively, in an embodiment, the control host **11** can also be distributed in the handling machines **12(1)** to **12(n)** to form a distributed control network, and the handling machines **12(1)** to **12(n)** can perform wireless communication with each other.

(8) The handling machines **12(1)** to **12(n)** can be used to move, lift up or put down specific objects. Taking the handling machine **12(1)** as an example, the handling machine **12(1)** may have a moving and rotating mechanism to perform actions such as going forward, going backward, and/or turning on the ground. In addition, the handling machine **12(1)** may have one or more extension arms. For example, the handling machine **12(1)** can control its extension arm to move vertically or horizontally. In this way, the handling machine **12(1)** can use an extension arm to handle specific objects. For example, the handling machine **12(1)** can be an object handling device such as a stacker, a pallet truck, or a forklift. The extension arm of the handling machine **12(1)** can be implemented as a mechanical component such as a clamp, a tray, or a mechanical arm that can be used to handle specific objects. Moreover, the handling machine **12(1)** can be configured to handle the object to be handled itself, or a conveyer such as a pallet, a tray, or a carrier used to carry the object to be handled.

(9) FIG. 2 is a functional block diagram of the control host according to an embodiment of the disclosure. Please refer to FIG. 2, the control host **11** includes a processor **21**, a storage circuit **22** and an input/output (I/O) interface **23**. The processor **21** can be configured to control the whole or part of the operation of the control host **11**. For example, the processor **21** may include a central processing unit (CPU), or other programmable general-purpose or special-purpose microprocessors, digital signal processors (DSP), programmable controller, application specific integrated circuits (ASIC), programmable logic device (PLD) or other similar devices or a combination of these devices. It should be noted that in the following embodiments, the description of the processor **21** is equivalent to the description of the control host **11**.

(10) The storage circuit **22** is coupled to the processor **21** and configured to store data. For example, the storage circuit **22** may include a volatile storage circuit and a non-volatile storage circuit. The volatile storage circuit is configured to volatily store data. For example, the volatile storage circuit may include a random access memory (RAM) or similar volatile storage media. The non-volatile storage circuit is configured for non-volatily storing data. For example, the non-volatile storage circuit may include a read only memory (ROM), a solid state disk (SSD), a conventional hard disk drive (HDD) or similar non-volatile storage media.

(11) The input/output interface **23** is coupled to the processor **21** and configured to transmit signals. For example, the processor **21** can receive input signals or transmit output signals through the input/output interface **23**. For example, the input/output interface **23** may include various input/output devices such as a wireless communication interface (such as a wireless network card), a mouse, a keyboard, a monitor, a touch panel, and/or a speaker.

(12) In an embodiment, the processor **21** can detect a handling instruction. This handling instruction can be used to instruct handling of a specific object (also referred to as the first object). For example, the processor **21** may receive a handling instruction input by an administrator through the input/output interface **23**. Alternatively, the processor **21** may automatically generate and detect the handling instruction according to a system schedule. In an embodiment, the processor **21** can determine the object to be handled (i.e., the first object) according to the handling instruction and try to read the handling information **201** of the first object from the storage circuit **22**.

(13) In an embodiment, if the handling information **201** can be read (for example, the handling information **201** is currently stored in the storage circuit **22** and can be read normally), the processor **21** can control, according to the handling information **201**, a certain handling machine (also referred to as the first handling machine) to move to a specific position (also referred to as the first position) and perform a specific operation (also referred to as a first operation) to lift up (including clamping or forklifting) the first object at the first position.

(14) In an embodiment, the first operation includes adjusting the posture of the first handling machine so that the first handling machine lifts up the first object based on a specific posture (also referred to as the first posture). For example, the step of adjusting the posture of the first handling machine includes adjusting the angle between the extension arm of the first handling machine and the first object, so that the extension arm of the first handling machine is at a position suitable for lifting up the first object. In this manner, with the first posture, the first handling machine can use the extension arm to lift up the first object smoothly. From another perspective, the first posture can reflect the posture of the first handling machine when lifting up the first object at the first position. For example, the first posture may reflect that the first handling machine lifts up the first object in a specific angle state (also referred to as the first angle state).

(15) FIG. 3 is a schematic diagram of adjusting the posture of the first handling machine according to an embodiment of the disclosure. Please refer to FIG. 1 to FIG. 3, assuming that the first handling machine is the handling machine **12(1)** and the first object is an object **13**. For example, the object **13** may be a pallet. After controlling the handling machine **12(1)** to move to the vicinity (i.e., the first position) of the position of the object **13**, the processor **21** can adjust the posture of the handling machine **12(1)** according to the handling information **201**. For example, the processor **21** can control the handling machine **12(1)** to rotate a specific angle Θ at the first position according to the handling information **201**, so that the extension arms **101** and **102** of the handling machine **12(1)** are aligned with the grooves **131** and **132** under the object **13**, respectively. After the handling machine **12(1)** is controlled to rotate by the angle Θ at the first position, it can be regarded that the handling machine **12(1)** is in the first posture. Under the circumstances, the processor **21** can control the handling machine **12(1)** to lift up the first object based on the first posture. For example, the processor **21** can control the handling machine **12(1)** to extend the extension arms **101** and **102** forward and insert them into the grooves **131** and **132**, respectively. After the extension arms **101** and **102** are inserted into the grooves **131** and **132**, respectively, the processor **21** can control the handling machine **12(1)** to lift up the object **13**.

(16) In an embodiment, after the first object is lifted up, the processor **21** may continue to control the first handling machine to move to another position (also referred to as the second position) and perform another operation (also referred to as second operation) to put down the first object at the second position. The second position is different from the first position.

(17) In an embodiment, the second operation includes adjusting the posture of the first handling machine so that the first handling machine puts down the first object based on another posture (also referred to as the second posture). For example, at the second position, the processor **21** may control the first handling machine to rotate another specific angle. Then, the processor **21** can control the first handling machine to put down the first object in a specific angle state (also referred to as a second angle state). In other words, the second posture can reflect the posture of the first handling machine when putting down the first object at the second position. For example, the second posture may reflect that the first handling machine puts down the first object in a specific angle state (i.e., the second angle state).

(18) In an embodiment, the processor **21** may detect the first posture and the second posture through a sensor arranged inside the first handling machine. For example, the sensor may include a sensor that senses the current posture (such as the angle state) of the handling machine such as a global positioning system (GPS) sensor, a G-sensor, an accelerometer and/or a gyroscope, etc.

(19) In an embodiment, in response to the first object being placed at the second position, the processor **21** may update the handling information **201**. For example, the processor **21** may update the handling information **201** from the current handling information (also referred to as the first handling information) to the new handling information (also referred to as the second handling information). The first handling information carries information related to the first posture. For example, the first handling information may carry the information of the first angle state (for example, the angle value of the first handling machine in the first posture). The second handling

information carries information related to the second posture. For example, the second handling information can carry the information of the second angle state (for example, the angle value of the first handling machine in the second posture). In an embodiment, the first handling information further carries information related to the first position. For example, the first handling information may carry the information of the first position (for example, the coordinates of the first position). In an embodiment, the second handling information further carries information related to the second position. For example, the second handling information may carry the information of the second position (for example, the coordinates of the second position).

(20) In an embodiment, after determining the object to be handled (i.e., the first object), the processor **21** can obtain the position of the first object (i.e., the first position) and the posture (that is, the first posture) to be used to lift up the first object according to the handling information **201**. In this way, the processor **21** can control, according to the handling information **201**, the first handling machine to move to the first position and lift up the first object based on the first posture. In addition, after the first object is put down at the second position, the processor **21** can update the information of the second position and the second posture to the handling information **201**. Thereafter, when the first object needs to be handled again, the processor **21** may control, according to the updated handling information **201**, another handling machine to move to the second position and lift up the first object again based on the second posture.

(21) In an embodiment, the first handling machine may have a camera modules. For example, the camera modules may include a two-dimensional (2D) camera and/or a three-dimensional (3D) camera. A two-dimensional camera can be used to capture two-dimensional images. A 3D camera can be used to capture 3D images. Compared with a two-dimensional image, a three-dimensional image can further carry depth information of the object (such as the first object) in the image. The depth information can be used to reflect the distance between the object and the camera modules (or the first handling machine). In an embodiment, the camera modules may also include various optical sensors and/or depth sensors such as lidar, and the disclosure provides no limitation to the type of the camera modules.

(22) In an embodiment, the first handling machine may not have a camera modules. In an embodiment, regardless of whether the first handling machine has a camera modules or not, according to the handling information **201**, the processor **21** can control the first handling machine according to the handling information **201** to move to the first position and perform the first operation to lift up the first object at the first position without performing image recognition on the first object by the first handling machine.

(23) In an embodiment, after detecting the handling instruction, if the handling information (for example, handling information **201**) of the first object cannot be read, the processor **21** may control another handling machine (also referred to as the second handling machine) to move to the first position and perform image recognition through the second handling machine to detect the first object at the first position. In particular, the second handling machine has a camera modules. The processor **21** can control the second handling machine to perform the first operation to lift up the first object according to the recognition result of the image recognition. After the first object is lifted up, the processor **21** can control the second handling machine to move to the second position and perform the second operation to put down the first object at the second position. In response to the first object being put down at the second position, the processor **21** may further update the handling information of the first object. For example, the processor **21** can update the information of the second position and the second posture to the handling information **201** and store the handling information **201** in the storage circuit **22**. Thereafter, the processor **21** may query the handling information **201** to automatically handle the first object.

(24) FIG. 4 is a schematic diagram of image recognition according to an embodiment of the disclosure. Referring to FIG. 4, in an embodiment, if the handling information (for example, handling information **201**) of the first object cannot be read, the processor **21** may control the

second handling machine to move to the first position where the first object is located and capture an image **41** through the camera modules of the second handling machine. The image **41** exhibits at least a part of the first object (i.e., the object **13**). The processor **21** can perform image recognition on the image **41** to detect a specific region **401** in the image **41**. The processor **21** can adjust the angle between the extension arm of the second handling machine and the first object according to the pixel distribution of the image in the specific region **401**, so that the extension arm of the second handling machine is in a position suitable for lifting up the first object. Then, the processor **21** can control the extension arm of the second handling machine to extend forward and lift up the first object. For details of related operations, refer to the embodiment in FIG. 3, and no repetition is incorporated herein. In addition, the image recognition can be performed by trained deep learning models, neural network models or similar artificial intelligence models.

(25) In an embodiment, the handling information of multiple objects to be handled can be stored in the storage circuit **22** of the control host **11** in a centralized manner. When it is necessary to query the handling information of a specific object, the control host **11** can control the specific handling machine to handle the specific object according to the handling information corresponding to the specific object in the storage circuit **22**. Or, in an embodiment, the handling information of multiple objects to be handled can also be shared between multiple handling machines (for example, handling machines **12(1)** to **12(n)** in FIG. 1) through a distributed network architecture.

(26) FIG. 5 is a flowchart of a handling machine control method according to an embodiment of the disclosure. Please refer to FIG. 5, in step **S501**, a handling instruction is detected, which is used to instruct the handling of the first object. In step **S502**, the handling information of the first object is read according to the handling instruction. In step **S503**, it is determined whether the handling information of the first object is read.

(27) If the handling information of the first object is read, in step **S504**, the first handling machine is controlled, according to the handling information of the first object, to move to the first position and perform the first operation to lift up the first object at the first position. After the first object is lifted up, in step **S505**, the first handling machine is controlled to move to the second position and perform the second operation to put down the first object at the second position. In step **S506**, in response to the first object being put down at the second position, the handling information of the first object is updated.

(28) On the other hand, if the handling information of the first object cannot be read, in step **S507**, the second handling machine is controlled, according to the handling information of the first object, to move to the first position and perform an image recognition to detect the first object at the first position through the second handling machine, and the second handling machine has a camera modules. In step **S508**, the second handling machine is controlled to perform the first operation to lift up the first object according to the recognition result of the image recognition. After the first object is lifted up, in step **S509**, the second handling machine is controlled to move to the second position and perform the second operation to put down the first object at the second position. Thereafter, in step **S506**, in response to the first object being put down at the second position, the handling information of the first object is updated.

(29) However, each step in FIG. 5 has been described in detail as above, and will not be repeated here. It should be noted that each step in FIG. 5 can be implemented as multiple program codes (such as a software module) or a circuit (such as a circuit module), and this disclosure provides no limitation thereto. In addition, the method of FIG. 5 can be used in conjunction with the above exemplary embodiments, or can be used alone, and this disclosure provides no limitation thereto.

(30) In summary, the handling machine control method, the handling machine control system, and the control host provided in this disclosure can record the handling information (such as position information and posture information) when the handling machine handles the first object and store this information for handling the first object next time. By automatically handling objects based on the handling information of the objects and continuously updating the handling information of the

objects, the entire handling system can achieve almost fully automated operation, thereby effectively improving the efficiency of automated handling of goods.

(31) Although the disclosure has been disclosed as above by embodiments, it is not intended to limit the disclosure. Anyone with ordinary knowledge in the technical field can make some changes and refinement without departing from the spirit and scope of the disclosure. Therefore, the scope to be protected by this disclosure shall be subject to the scope of the attached claims.

Claims

1. A handling machine control method, adaptable for a handling machine control system, wherein the handling machine control method comprises: detecting a handling instruction which instructs a handling of a first object; reading handling information of the first object according to the handling instruction; before controlling any one of a first handling machine and a second handling machine to move the first object, determining whether the handling information of the first object can be read, wherein the first handling machine does not have any camera module, and the second handling machine has a camera module; in response to a determination result reflecting that the handling information of the first object is read, controlling, according to the handling information of the first object, the first handling machine to move to a first position and perform a first operation to lift up the first object at the first position; after the first object is lifted up by the first handling machine, controlling the first handling machine to move to a second position and perform a second operation to put down the first object at the second position; in response to a determination result reflecting that the handling information of the first object cannot be read, controlling the second handling machine to move to the first position and perform an image recognition, by the second handling machine, to detect a specific region in an image captured by the camera module; adjusting an angle between an extension arm of the second handling machine and the first object according to a pixel distribution of the image in the specific region and then controlling the extension arm of the second handling machine to extend forward and lift up the first object; after the first object is lifted up by the second handling machine, controlling the second handling machine to move to the second position and perform a second operation to put down the first object at the second position; and in response to that the first object is put down at the second position, updating the handling information of the first object.

2. The handling machine control method according to claim 1, wherein the first operation comprises adjusting a posture of the first handling machine so that the first handling machine lifts up the first object based on a first posture, and the second operation comprises adjusting the posture of the first handling machine so that the first handling machine puts down the first object based on a second posture.

3. The handling machine control method according to claim 2, wherein the step of updating the handling information of the first object comprises: updating the handling information of the first object from first handling information to second handling information, wherein the first handling information carries information related to the first posture, and the second handling information carries information related to the second posture.

4. The handling machine control method according to claim 1, wherein the step of controlling, according to the handling information of the first object, the first handling machine to move to the first position and perform the first operation to lift up the first object at the first position comprises: controlling the first handling machine according to the handling information of the first object to move to the first position and perform the first operation to lift up the first object at the first position without performing an image recognition on the first object by the first handling machine.

5. A handling machine control system, comprising: a processor; a second handling machine, coupled to the processor; and a first handling machine, coupled to the processor, wherein the processor is configured to: detect a handling instruction which instructs a handling of a first object;

read handling information of the first object according to the handling instruction; before controlling any one of the first handling machine and the second handling machine to move the first object, determine whether the handling information of the first object can be read, wherein the first handling machine does not have any camera module, and the second handling machine has a camera module; in response to a determination result reflecting that the handling information of the first object is read, control, according to the handling information of the first object, the first handling machine to move to a first position and perform a first operation to lift up the first object at the first position; after the first object is lifted up by the first handling machine, control the first handling machine to move to a second position and perform a second operation to put down the first object at the second position; in response to a determination result reflecting that the handling information of the first object cannot be read, control the second handling machine to move to the first position and perform an image recognition, by the second handling machine, to detect a specific region in an image captured by the camera module; adjust an angle between an extension arm of the second handling machine and the first object according to a pixel distribution of the image in the specific region and then control the second handling machine to extend forward and lift up the first object; after the first object is lifted up by the second handling machine, control the second handling machine to move to the second position and perform a second operation to put down the first object at the second position; and in response to that the first object is put down at the second position, update the handling information of the first object.

6. The handling machine control system according to claim 5, wherein the first operation comprises adjusting a posture of the first handling machine so that the first handling machine lifts up the first object based on a first posture, and the second operation comprises adjusting the posture of the first handling machine so that the first handling machine puts down the first object based on a second posture.

7. The handling machine control system according to claim 6, wherein the operation of updating the handling information of the first object comprises: updating the handling information of the first object from first handling information to second handling information, wherein the first handling information carries information related to the first posture, and the second handling information carries information related to the second posture.

8. The handling machine control system according to claim 5, wherein the operation of controlling, according to the handling information of the first object, the first handling machine to move to the first position and perform the first operation to lift up the first object at the first position comprises: controlling the first handling machine according to the handling information of the first object to move to the first position and perform the first operation to lift up the first object at the first position without performing an image recognition on the first object by the first handling machine.

9. A control host for controlling at least one handling machine, and the control host comprising: a storage circuit, configured for storing handling information of a first object; and a processor, coupled to the storage circuit, wherein the processor is configured to: detect a handling instruction which instructs a handling of the first object; read the handling information of the first object according to the handling instruction; before controlling any one of a first handling machine and a second handling machine to move the first object, determine whether the handling information of the first object can be read, wherein the first handling machine does not have any camera module, and the second handling machine has a camera module; in response to a determination result reflecting that the handling information of the first object is read, control, according to the handling information of the first object, the first handling machine among the at least one handling machine to move to a first position and perform a first operation to lift up the first object at the first position; after the first object is lifted up by the first handling machine, control the first handling machine to move to a second position and perform a second operation to put down the first object at the second position; in response to a determination result reflecting that the handling information of the first object cannot be read, control a second handling machine among the at least one handling machine

to move to the first position and perform an image recognition, by the second handling machine, to detect a specific region in an image captured by the camera module; adjust an angle between an extension arm of the second handling machine and the first object according to a pixel distribution of the image in the specific region and then control the second handling machine to extend forward and lift up the first object; after the first object is lifted up by the second handling machine, control the second handling machine to move to the second position and perform a second operation to put down the first object at the second position; and in response to that the first object is put down at the second position, update the handling information of the first object.

10. The control host according to claim 9, wherein the first operation comprises adjusting a posture of the first handling machine so that the first handling machine lifts up the first object based on a first posture, and the second operation comprises adjusting the posture of the first handling machine so that the first handling machine puts down the first object based on a second posture.

11. The control host according to claim 10, wherein the operation of updating the handling information of the first object comprises: updating the handling information of the first object from first handling information to second handling information, wherein the first handling information carries information related to the first posture, and the second handling information carries information related to the second posture.

12. The control host according to claim 9, wherein the operation of controlling, according to the handling information of the first object, the first handling machine to move to the first position and perform the first operation to lift up the first object at the first position comprises: controlling the first handling machine according to the handling information of the first object to move to the first position and perform the first operation to lift up the first object at the first position without performing an image recognition on the first object by the first handling machine.
